CODE SPORT

Sandya Mannarswamy

In this month's column, we continue our discussion on content moderation on social media, turning our attention to misinformation detection and fact-checking of the content on the Web.

In last month's column, we had discussed some of the practical challenges associated with detecting offensive content in social media posts. Moderating online user generated content spans a wide spectrum of tasks, ranging from flagging offensive content to detecting misinformation or fake news on the Web.

Detecting misinformation is a difficult task, made much tougher by the fact that information on the Web is getting generated at an exponential scale. While there has never been a time in our history when knowledge and information has been so freely accessible to all, there has also never been such a deluge of misinformation, weaving a web of deceit, half-truths and falsehoods via the Internet. Unlike in the days when access to information was the major challenge, today, we consume an almost unparalleled volume of information. Our current challenge is to identify the lies, false claims and half-truths that hide themselves among the facts. Major world events such as Brexit, the 2016 US presidential elections, and the immigrant exodus into Europe have highlighted the explosive power of misinformation and fake news in leading the public astray.

While it has been the time-honoured duty of journalists to ensure that the public is presented with the facts and not misinformation, the Internet's information explosion has made it impossible for any manual fact-checking efforts to be effective, scalable and timely. Hence there has been considerable research in recent years that focuses on automating fact-checking of publicly available information. The automated fact-checking pipeline has two major phases:

- Identifying statements/sentences that need to be fact-checked. This is typically known as the claim identification phase.
- Verifying the identified claims using external knowledge sources. This is typically known as the claim verification phase.

Considering that the textual content on the Internet is of the order of exabytes, fact-checking each and every statement/sentence is a Herculean task even with state-of-art computing resources. An earlier study found that it takes only 2.5 hours for a meme to move from being reported in the news to becoming a part of Internet blogs. Given the rapid traversal of information across the Internet and that false news spreads much faster than true information, fact-checking needs to be timely for it to be useful.

Since fact-checking requires considerable computing resources, distinguishing between text containing a claim and text which simply expresses an opinion is critical for the success of any automated fact-checking system. Claim identification, which involves classifying which sentences in a text are potential claims that need to be verified, is an important task in the fight against misinformation and fake news.

Given a set of sentences, claim identification boils down to classifying each sentence as a potential claim or not. This brings up the question of, "What constitutes a claim?" Statements or utterances that are factual and objective are potential claims that need to be verified, whereas subjective opinions should not be checked. For example, consider the following two statements:

S1: Trump owns properties in India worth US$ 500 million.

S2: Trump is the worst president America has ever had.

Statement S1 is an objective statement, whereas S2 is a subjective statement. Hence we only need to consider S1 for claim verification whereas S2 need not be fact-checked.

While a statement needs to be objective or about facts to be considered a claim, not all factual sounding and objective statements need to be fact-checked. Only if the statement contains some information, which would be of interest to the public, does it need to go through the rigorous process of fact-checking.